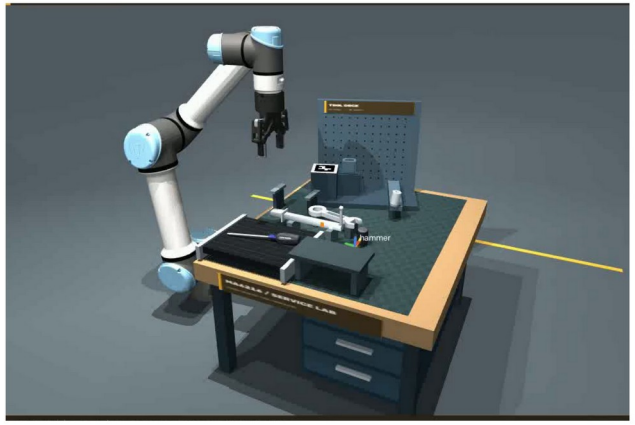
Calibrated 6D vision for sequential tool manipulation

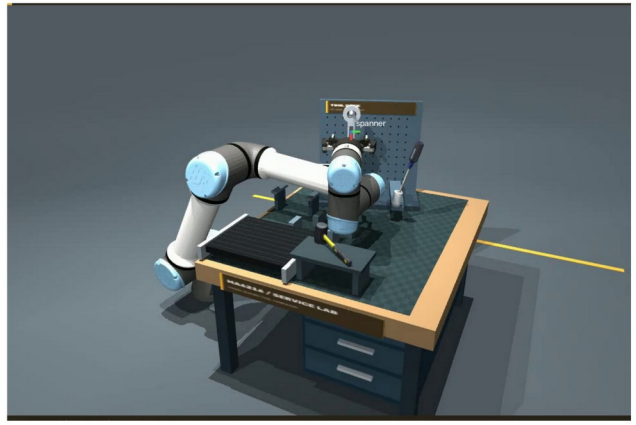
Problem & scope

a Initial workcell



t = 0.00 s | object pickup stations

b Released final state



t = 195.00 s | four acceptance checks passed

Study objective

UR5e + Robotiq 2F-85 in MuJoCo

RGB-D reconstruction + FoundationPose

Does execution-period pose feedback improve complete-task reliability?

Operation	Required relationship	Acceptance
Hammer return	Tool-to-tray pose	Released support
Moving screwdriver	Shaft-to-tilted-socket axis	Insertion + release
Independent rack	Frame-to-base seating	Stable fixture
Spanner regrasp / hang	New grasp; hole-to-pin alignment	Released hanging

Scope: one continuous four-operation scene. Contribution: independent integration and evaluation of existing methods.

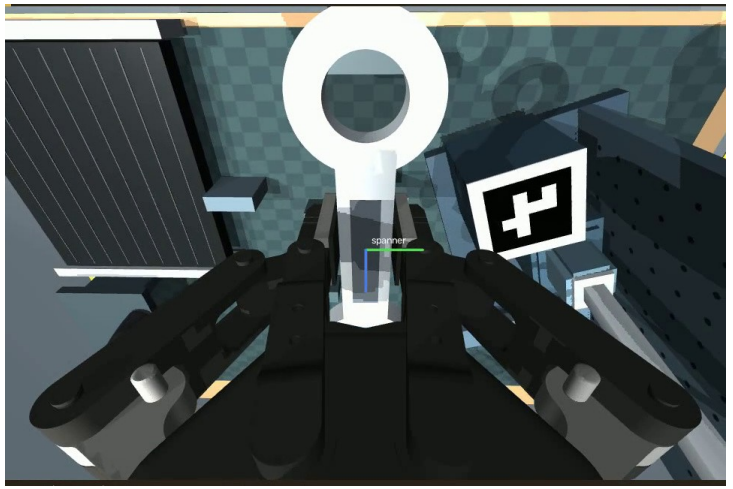
System architecture and technical selection

Methods

RGB-D
Estimated K
→
Localization
Depth foreground
→
6D pose
FoundationPose
→
Robot target
Calibrated chain
→
Execution
IK + control

Reobserve → update pose → bounded correction; simulation truth enters downstream scoring only.

Method / component	Output / role	Selection boundary
FoundationPose	Object-frame 6D pose	Selected; reconstructed RGB-D references
MegaPose	Model-based 6D pose	Reviewed; no external benchmark
GraspNet	Grasp candidates	Does not alone specify assembly pose
MuJoCo / OpenCV	Contact simulation / calibration	Reuse platform; inspectable interfaces



Model-free route: 24 calibrated RGB-D reference views → object-specific reconstruction → registration / tracking.

References and pretrained networks are still required; synthetic reference calibration remains an assumption.

Camera calibration: estimation and independent validation

Methods

a Checkerboard observations

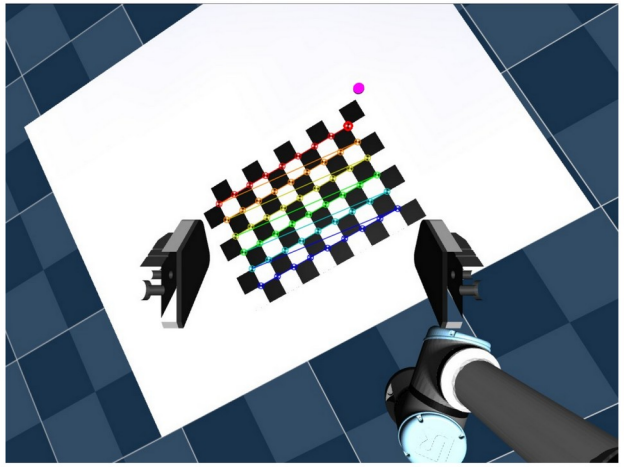
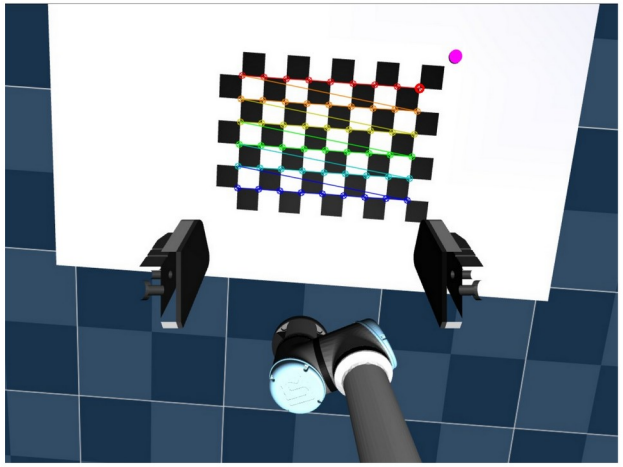


Image-based corner detection

b Held-out hand–eye view



PnP + robot FK; frozen validation chain

Procedure	Train / validation	Validation result
Wrist intrinsics	24 / 8 views	0.02177 px RMSE
Wrist hand–eye	24 / 10 views	0.852 mm / 0.0227°
Frozen chain	K, X and Y fixed	0.05497 px

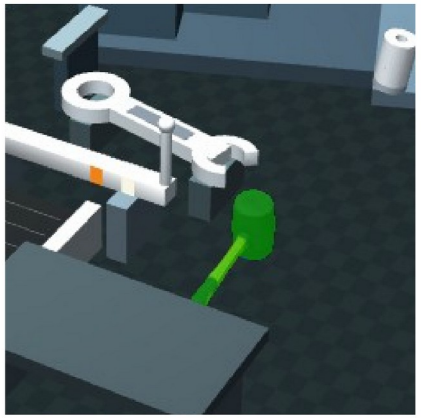
Estimation protocol

Checkerboard pixels → intrinsic calibration → image PnP + robot FK → AX = XB → training-only refinement.

Simulator K and camera pose are validation references only. Rejected images and held-out views are retained.
The known distortion-free pinhole model explains the small residuals; these are not real-camera accuracy claims.

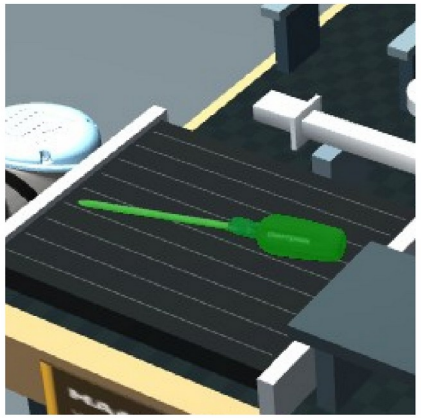
Localization: image masks, evaluation unit and assumptions

a Hammer



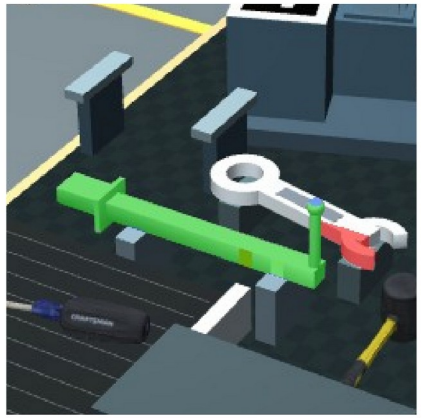
Nominal scene | seed 500

b Screwdriver



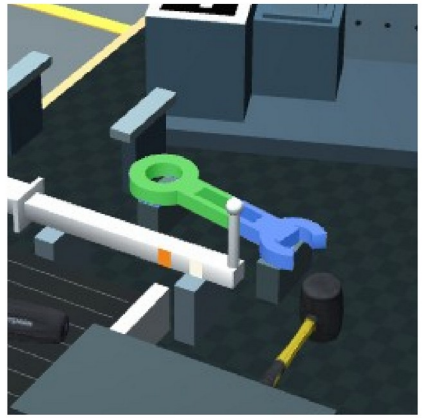
Nominal scene | seed 500

c Rack frame



Nominal scene | seed 500

d Spanner



Nominal scene | seed 500

Detector	Independent evaluation	Boundary
Depth foreground subtraction Known pickup regions	50 scenes / 200 queries 10 scenes per tool–condition cell	Matched empty backgrounds Restricted occlusion / noise
Predictions saved first	160/160 positive queries: IoU ≥ 0.5 0/40 absent-target false positives	Wilson and scene-bootstrap intervals Not 200 robot executions

Nominal seed 500. Green: overlap; blue: missed visible target; red: extra foreground. Crops preserve original overlays.

6D pose estimation and coordinate transformation

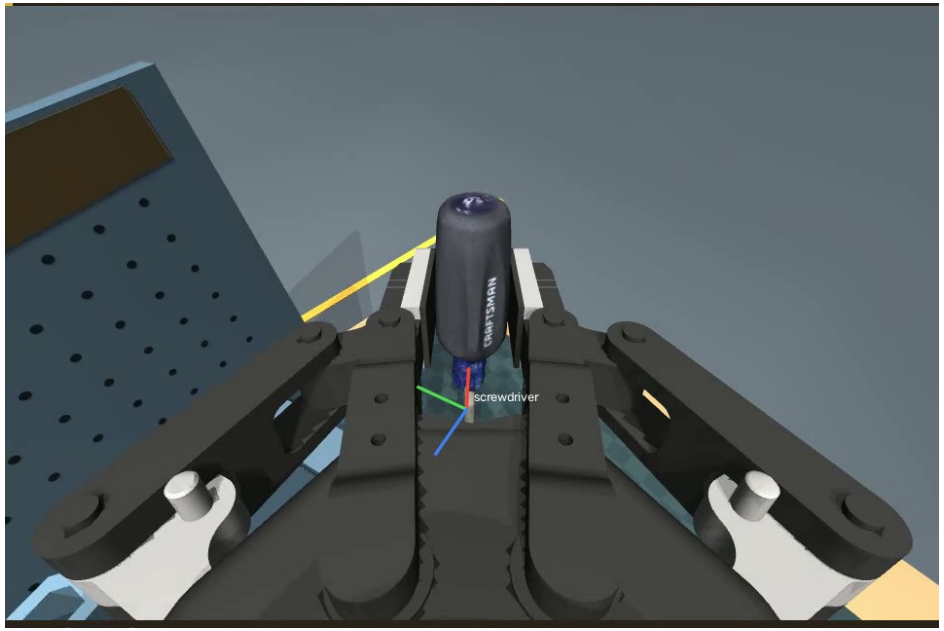
$$T_{BO} = T_{BG} \cdot T_{GC} \cdot T_{CO}$$

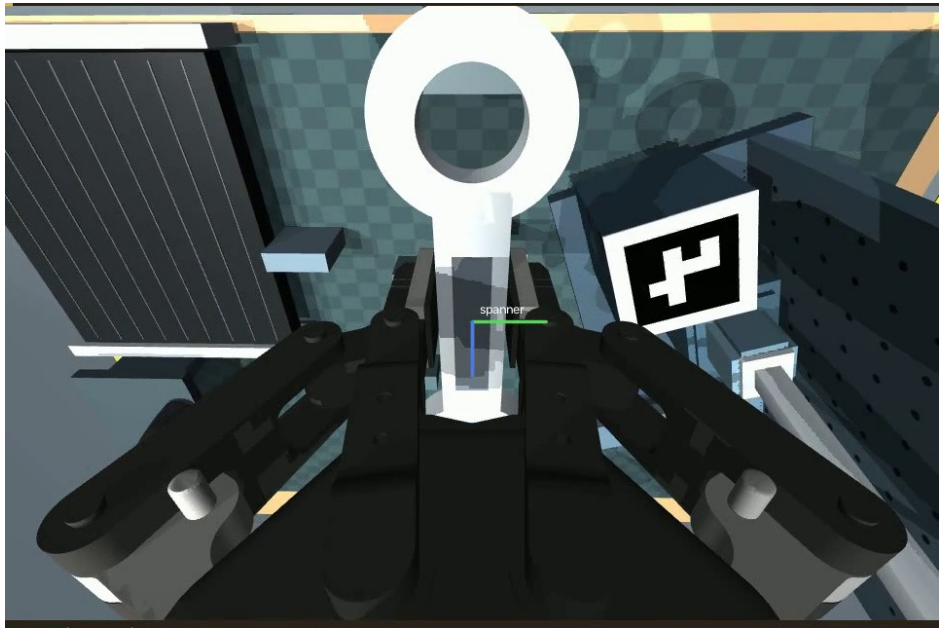
B: robot base G: gripper C: wrist camera O: object

T_AB maps coordinates from B into A.

FK supplies T_BG; calibration estimates T_GC.

FoundationPose supplies T_CO from visual observations.





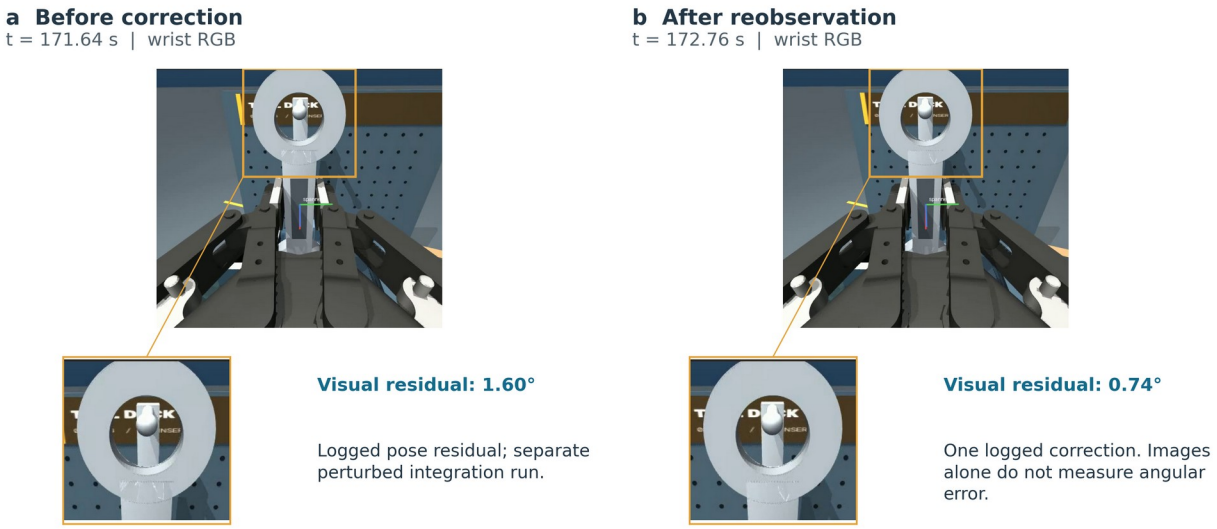
Task target: estimated object pose + task-specific geometric offset.

Regrasp changes the object-to-gripper relationship; observe again before generating the next target.

Visual feedback: modes, residuals and correction policy

Methods

Mode	Execution-period observation	Common components
Open loop	Initial acquisition / feed prediction only	Same estimator, calibration and guards
Staged	Selected task-transition reobservations	Same physical acceptance
Iterative	Repeated precontact pose correction	Staged observations retained



$$e_p = p_{des} - p_{est}$$

$$\Delta R = R_{des}\, R_{est}^{T}$$

Cap: 4 mm / 2° per step

Converged: two observations within 1.5 mm / 1.5°

Discrete precontact control; no stability proof.

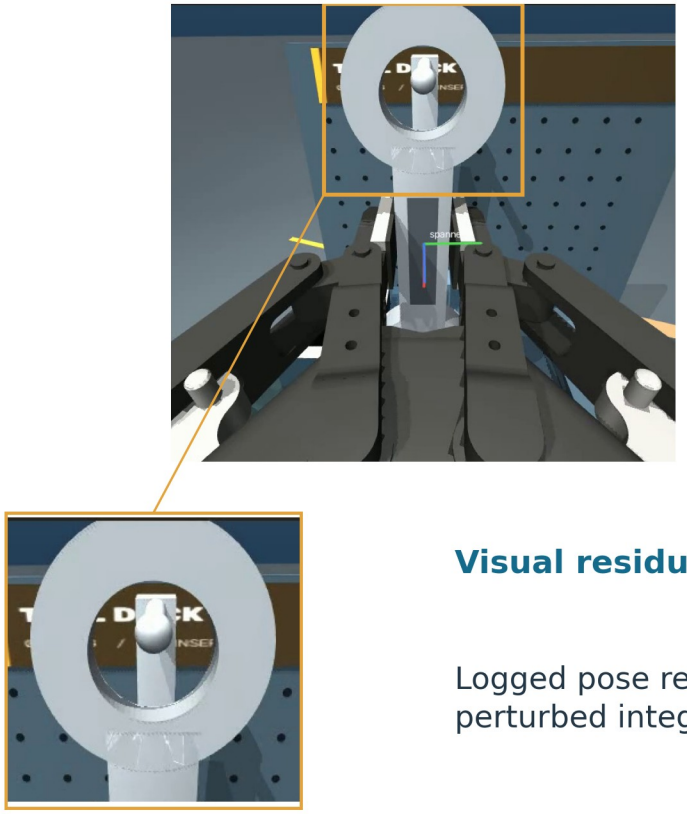
Correction images: separate perturbed integration run, one logged correction; 1.60° → 0.74° are visual residuals, not true geometric errors.

Visual correction: matched detail views before and after action

Separate integration diagnosis

a Before correction

t = 171.64 s | wrist RGB

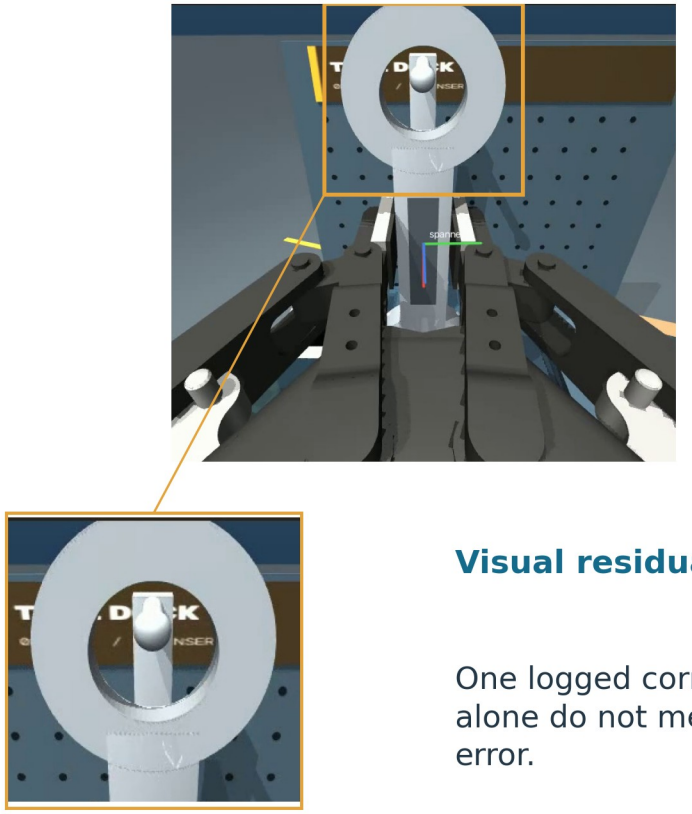


Visual residual: 1.60°

Logged pose residual; separate perturbed integration run.

b After reobservation

t = 172.76 s | wrist RGB



Visual residual: 0.74°

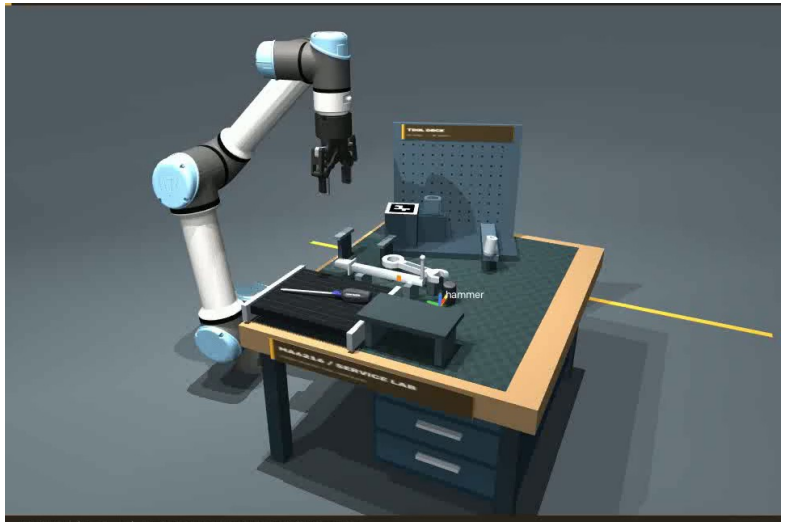
One logged correction. Images alone do not measure angular error.

One logged correction in a separate perturbed run. 1.60° and 0.74° are estimated angular residuals, not ground-truth errors.

Experimental design and metric definitions

Evaluation protocol

Factor	Levels / unit	Control
Feedback	Open loop / staged / iterative	Same pose estimator and camera calibration
Disturbance	None / alignment force–torque pulse	Delivery logged; early failures retained
Randomization	8 seeds per cell; 48 attempts	Initial states paired within seed
Primary outcome	All four final physical checks	Geometry + support + release + stability



Statistical reporting

Success: Wilson 95% confidence intervals.

Paired comparisons: exact tests with Holm adjustment.

Continuous metrics: trial-level summaries and bootstrap intervals.

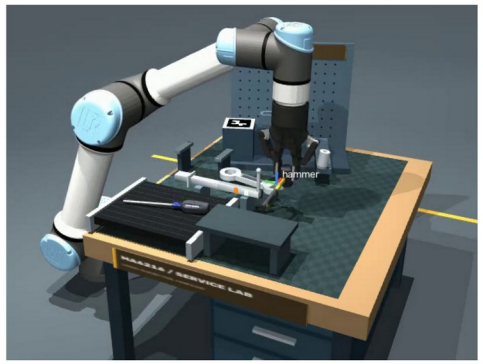
Seeds 400–407 are reused: paired replication, not a new holdout.

No task failures are removed or pooled with historical studies.

Execution evidence: hammer return and rack installation

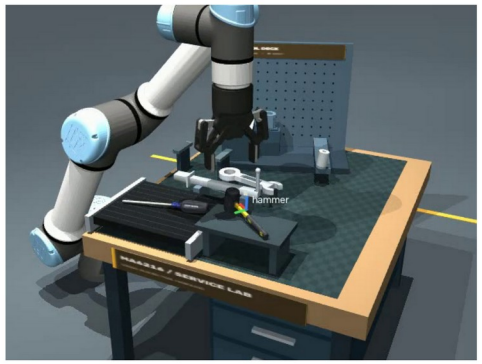
Representative successful run

a Lift and transport



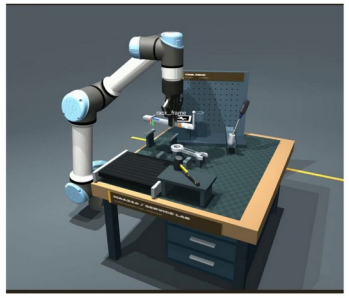
t = 13.60 s

b Released tray placement



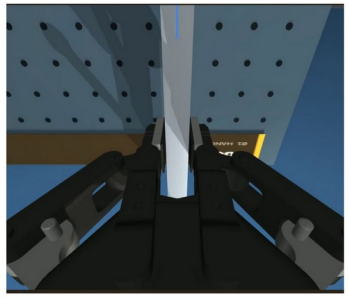
t = 25.00 s

a Independent frame pickup



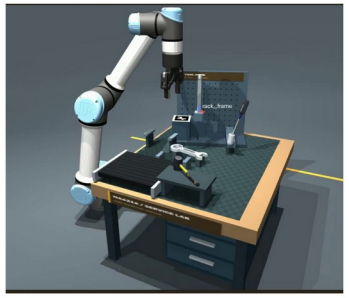
t = 91.00 s

b Frame–base seating



t = 112.00 s

c Released fixture



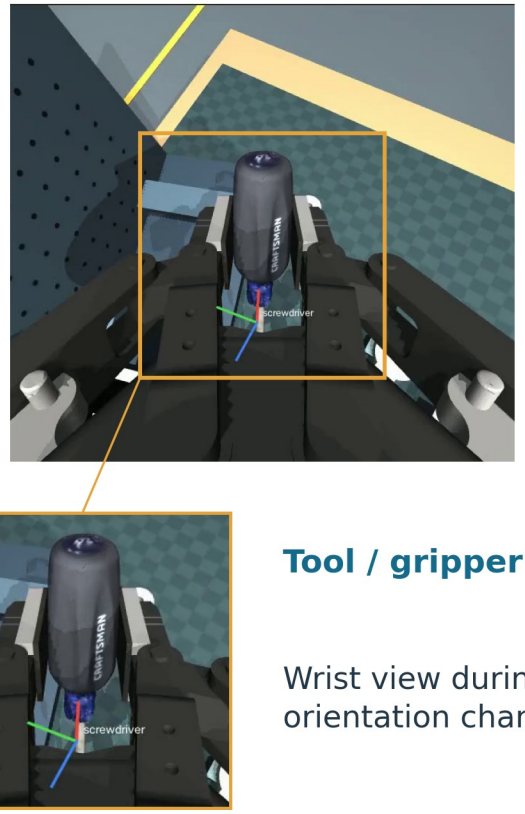
t = 134.40 s

Continuous run: hammer release precedes independent rack installation. Time labels refer to the original replay.

Tilted insertion: orientation, contact and released placement

Representative successful run

a Reorient
t = 50.00 s | wrist RGB

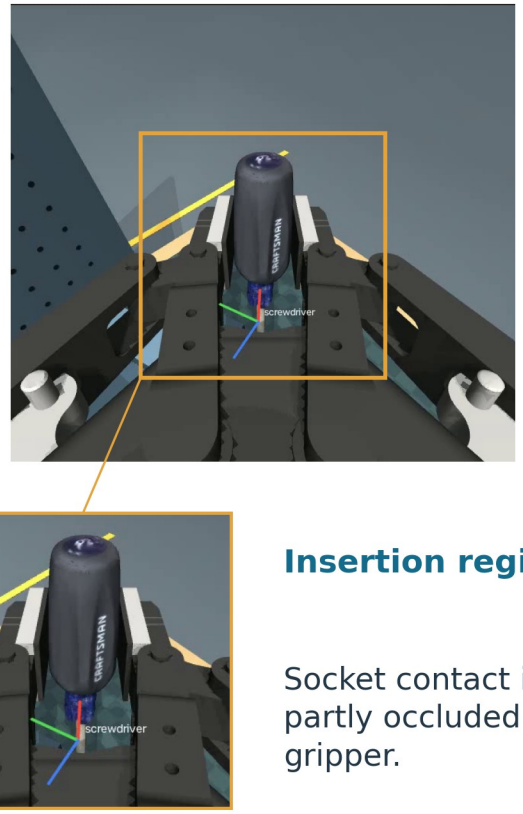




Tool / gripper

Wrist view during orientation change.

b Insert
t = 60.00 s | wrist RGB

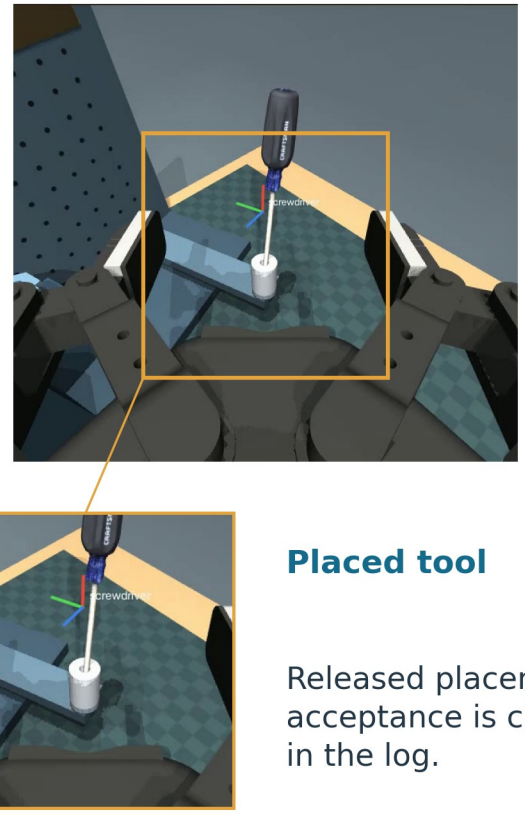


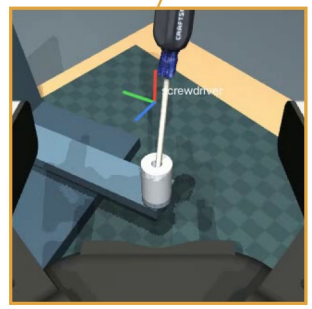


Insertion region

Socket contact is partly occluded by the gripper.

c Release
t = 70.00 s | wrist RGB





Placed tool

Released placement; acceptance is checked in the log.

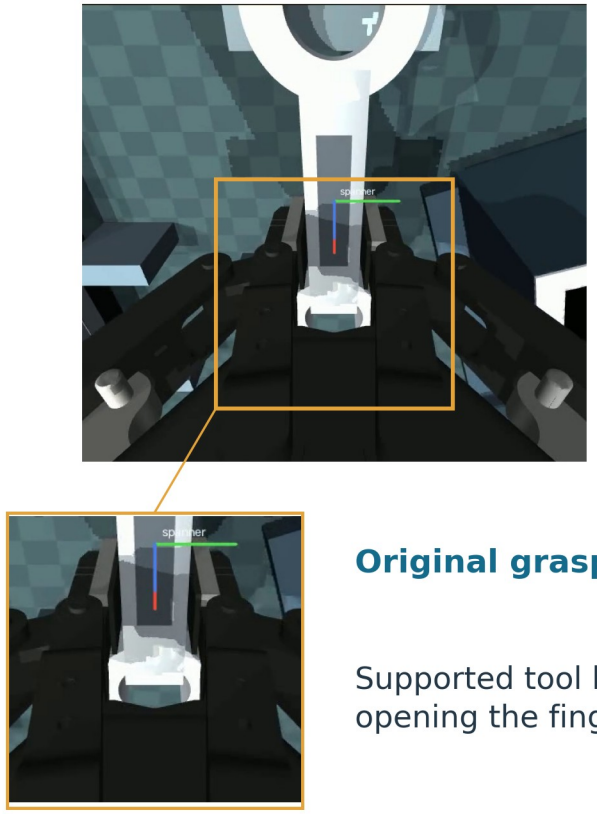
Connected insets show identical pixel windows. Contact is partly occluded; physical acceptance comes from the recorded checks.

Physical regrasp: support, clearance and grasp reacquisition

Representative successful run

a Place on support

t = 147.00 s | wrist RGB

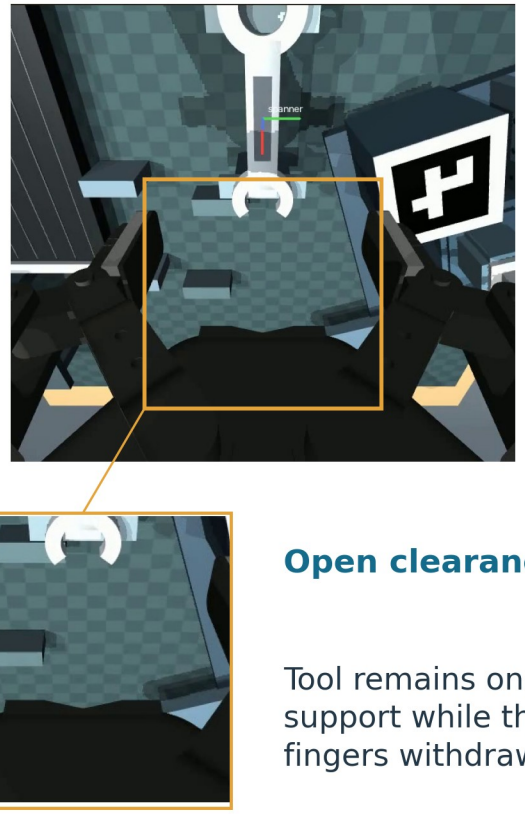


Original grasp

Supported tool before opening the fingers.

b Clear fingers

t = 150.40 s | wrist RGB

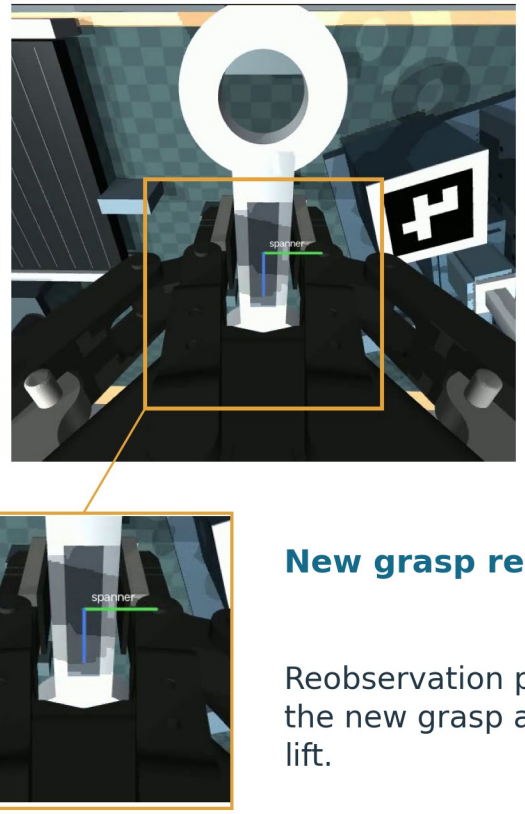


Open clearance

Tool remains on support while the fingers withdraw.

c Acquire new grasp

t = 158.20 s | wrist RGB



New grasp region

Reobservation precedes the new grasp and lift.

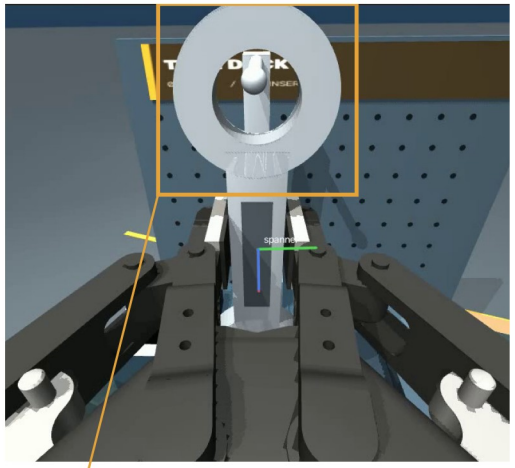
One continuous trajectory. Amber regions identify the grasp interface; new pose observation precedes reacquisition.

Hole-over-pin placement: local geometry and final support

Representative successful run

a Align

t = 171.00 s | wrist RGB



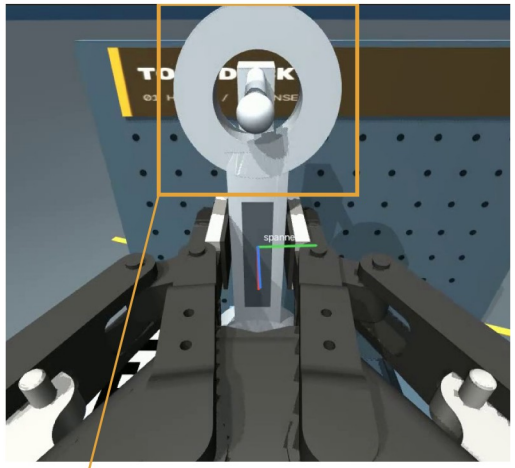


Ring / pin interface

Precontact view; the hole and pin are visible.

b Thread

t = 179.00 s | wrist RGB



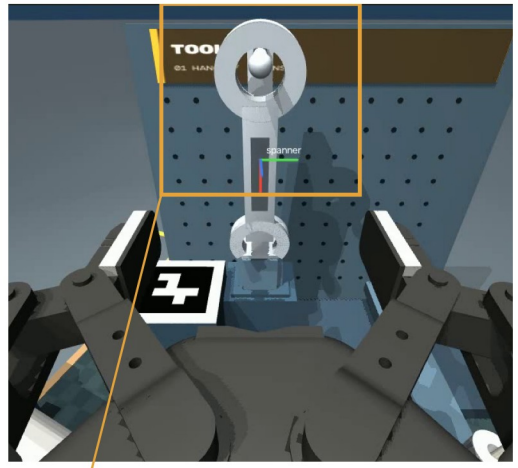


Pin inside ring

The enlarged view exposes the local relationship.

c Release

t = 195.00 s | wrist RGB





Released support

Gripper withdrawn; terminal checks pass in the log.

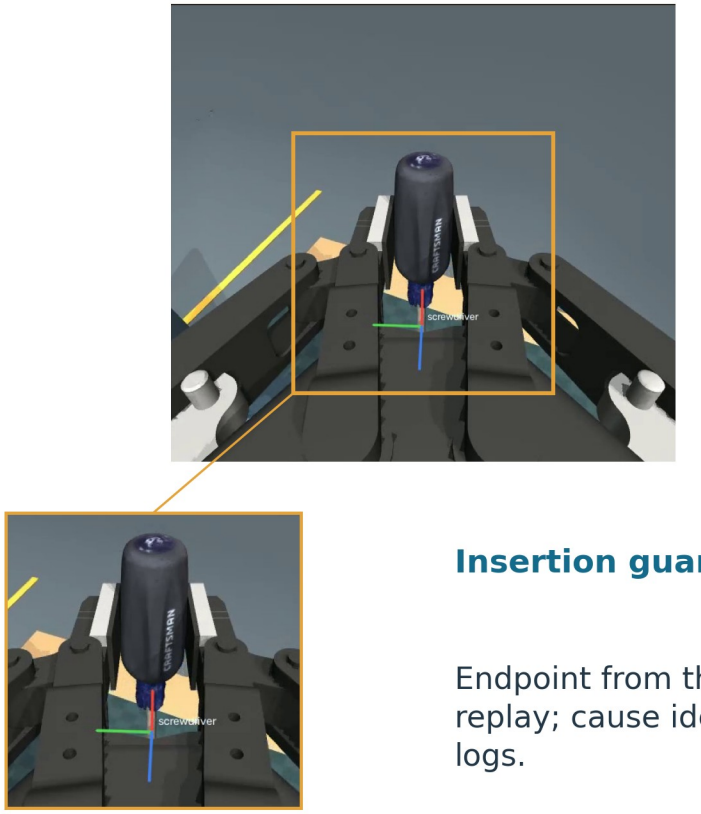
Four terminal checks passed. Separate passive-stability audit: 30 s. The insets explain geometry, not measured contact forces.

Matched-seed comparison: insertion stop versus released placement

Controlled examples

a Open loop / seed 400

t = 57.96 s | wrist RGB

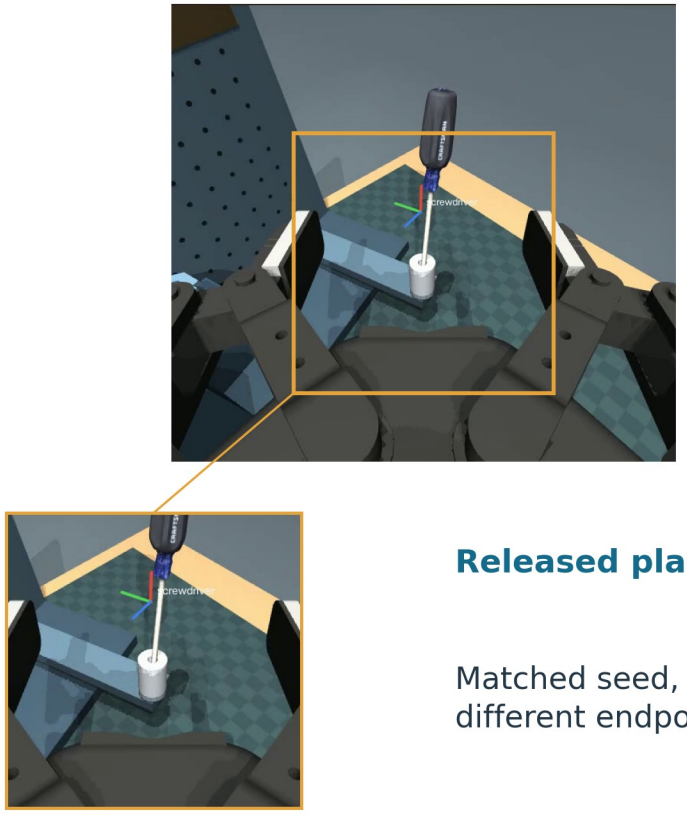


Insertion guard stop

Endpoint from the failure replay; cause identified from logs.

b Iterative / seed 400

t = 70.00 s | wrist RGB



Released placement

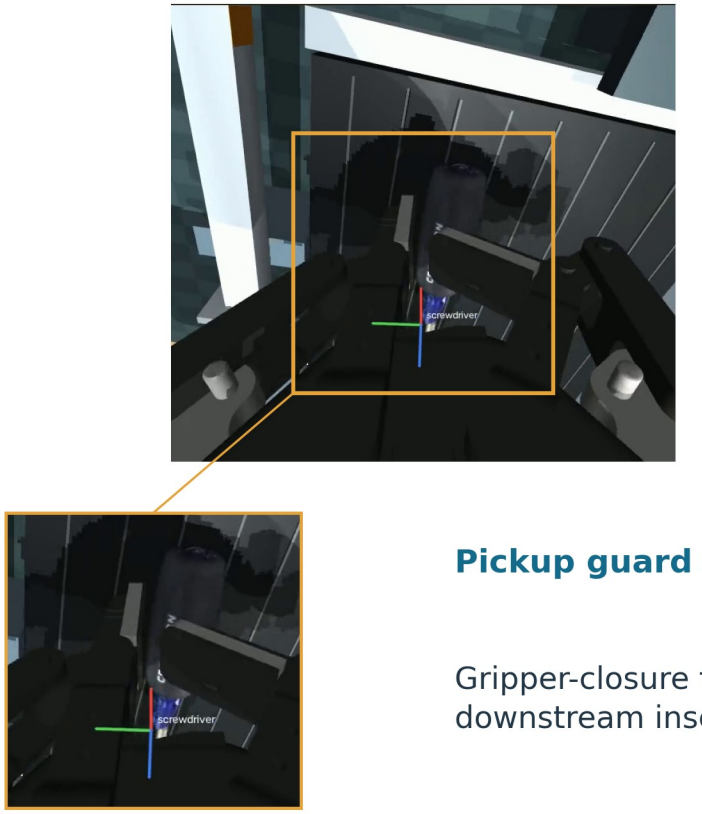
Matched seed, no disturbance; different endpoint times.

Seed 400, no disturbance; endpoint times differ. A selected paired example does not establish statistical superiority.

Shared failure boundary: gripper closure before insertion

Controlled examples

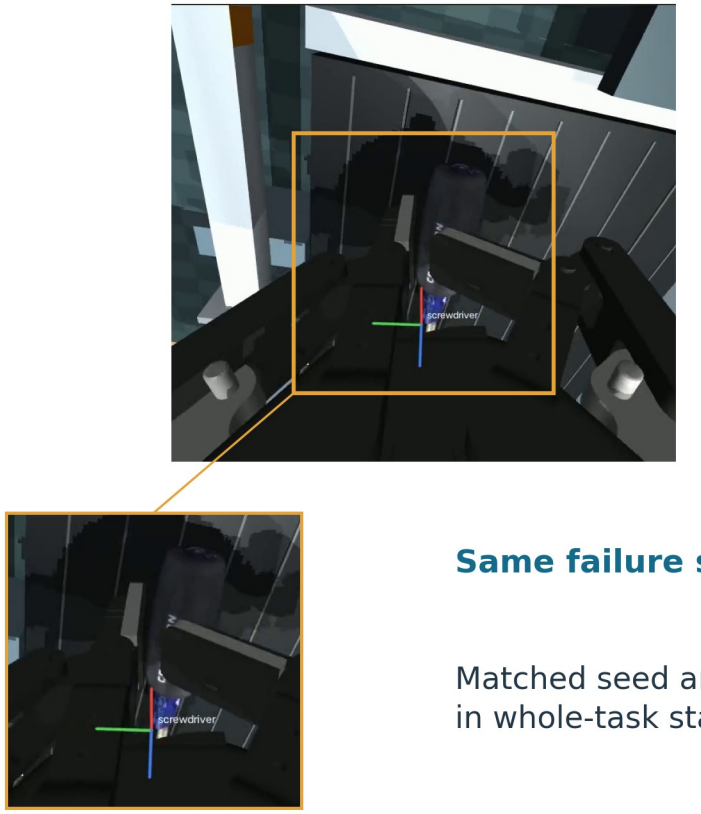
a Staged / seed 401
t = 40.24 s | wrist RGB



Pickup guard stop

Gripper-closure failure before downstream insertion control.

b Iterative / seed 401
t = 40.24 s | wrist RGB



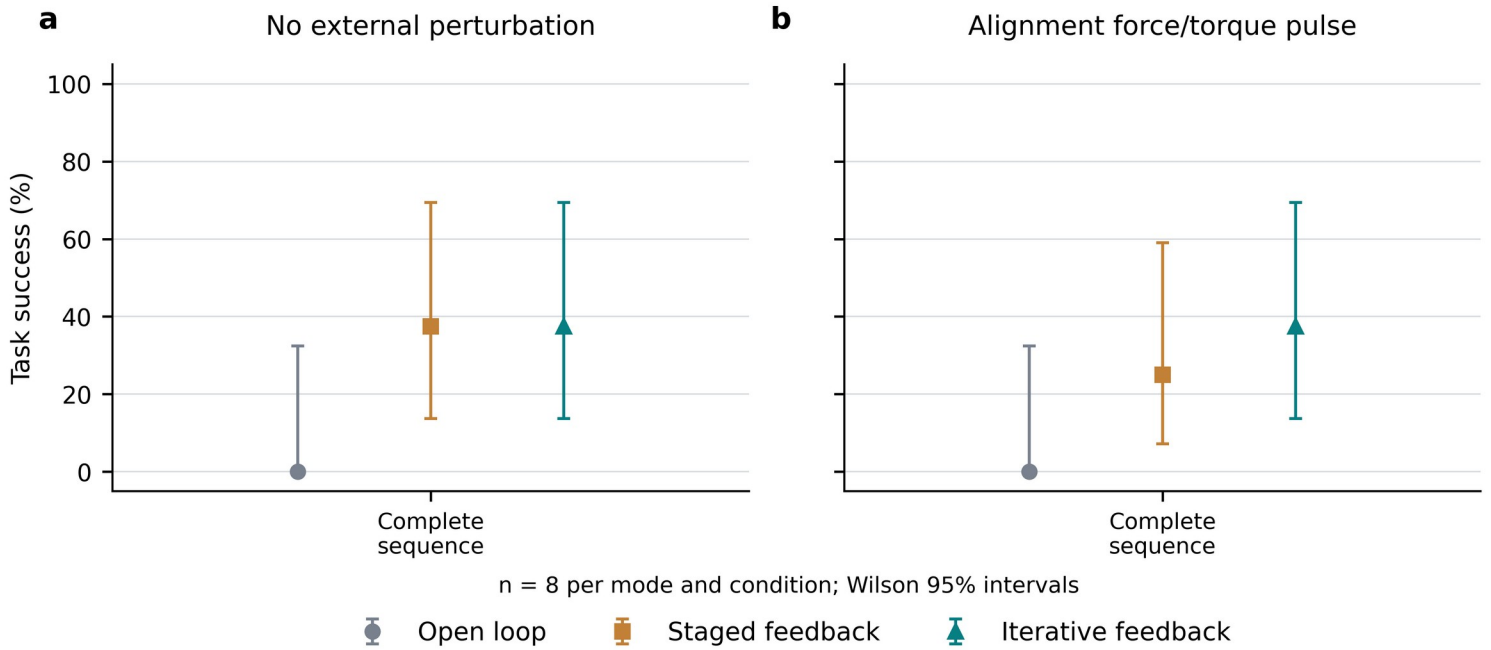
Same failure stage

Matched seed and stage; retained in whole-task statistics.

Seed 401: both modes stop at gripper closure. Stage and guard attribution are from logs; all failed attempts remain in the analysis.

Whole-sequence results and statistical uncertainty

Quantitative results



Mode	None	Pulse
Open loop	0/8	0/8
Staged	3/8	2/8
Iterative	3/8	3/8

11/48 complete successes.
All four adjusted paired p-values: 1.0.
No statistically established superiority or equivalence in this sample.

Interpretation

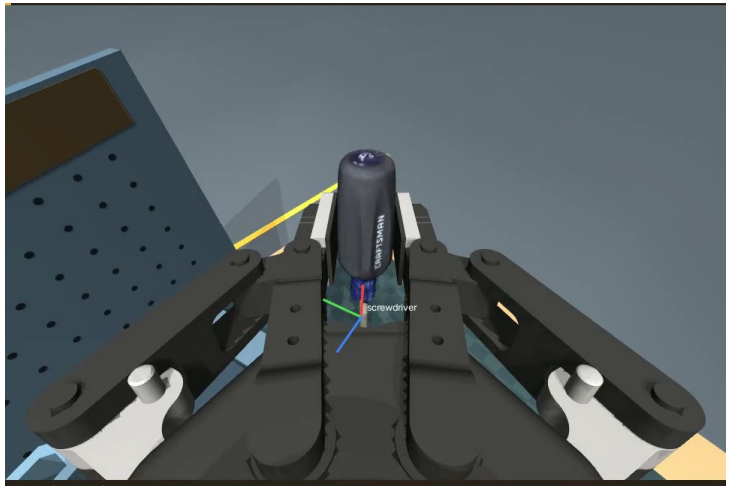
The selected successful trajectory demonstrates feasibility. Low complete-sequence reliability and wide intervals limit conclusions about feedback benefit. Early failures remain part of the system-level outcome.

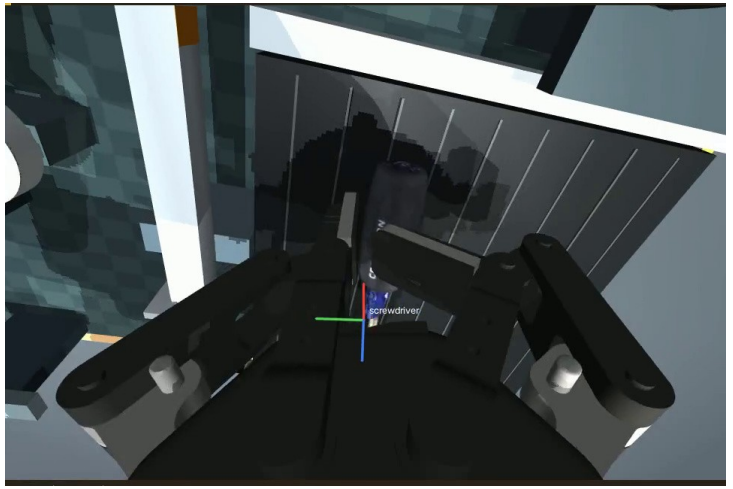
Diagnostic results: pose accuracy and physical failures

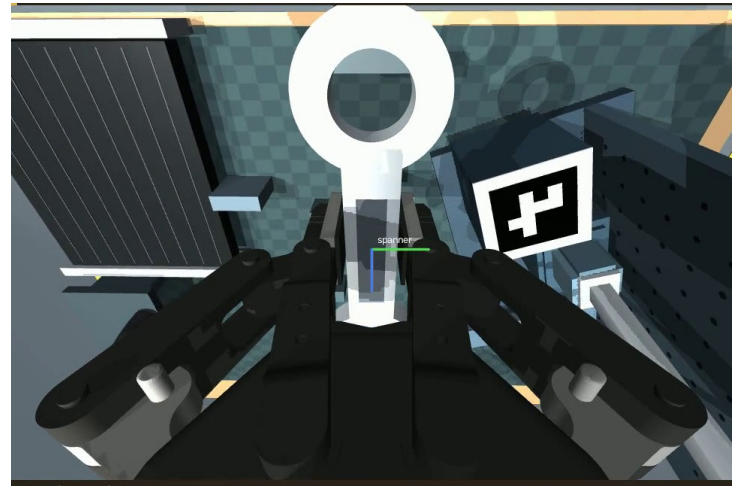
Error decomposition

Initial pose metric	Mean
Translation error	2.60 mm
Rotation error	1.44°
ADD / ADD-S	3.17 / 2.29 mm

Failure category	Attempts
Moving-pickup contact guard	24
Hammer approach planning	6
Open-loop insertion guard	6
Bilateral spanner grasp loss	1



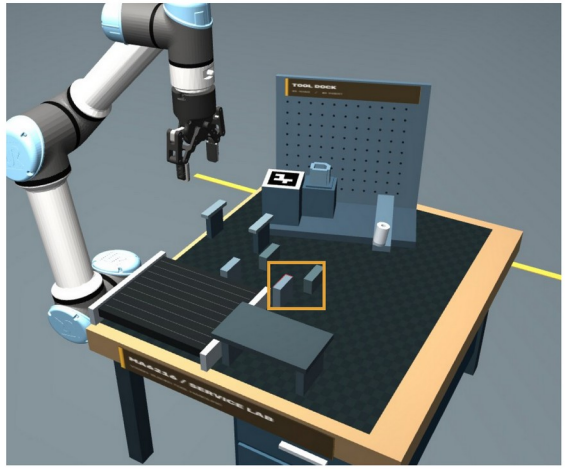




Different endpoints: initial pose error, TCP tracking error, final placement and task success. 500 Hz physics is not visual feedback frequency.

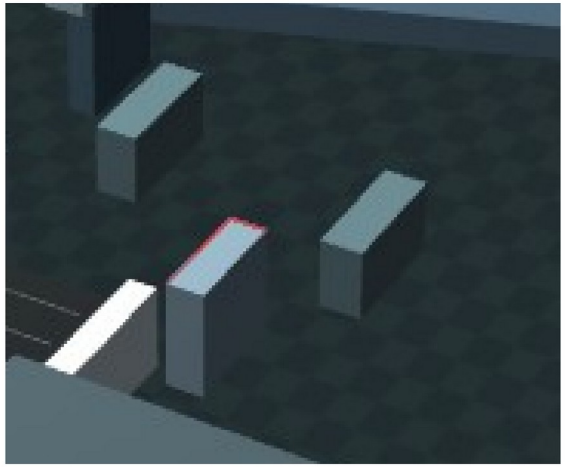
Generalization boundary: background-dependent localization

a Target-absent scene



Separate background-mismatch diagnosis

b Magnified false-positive region



Red pixels: predicted foreground without target

Observed limitation

Absent rack target, but mismatched background geometry produces a false-positive region.

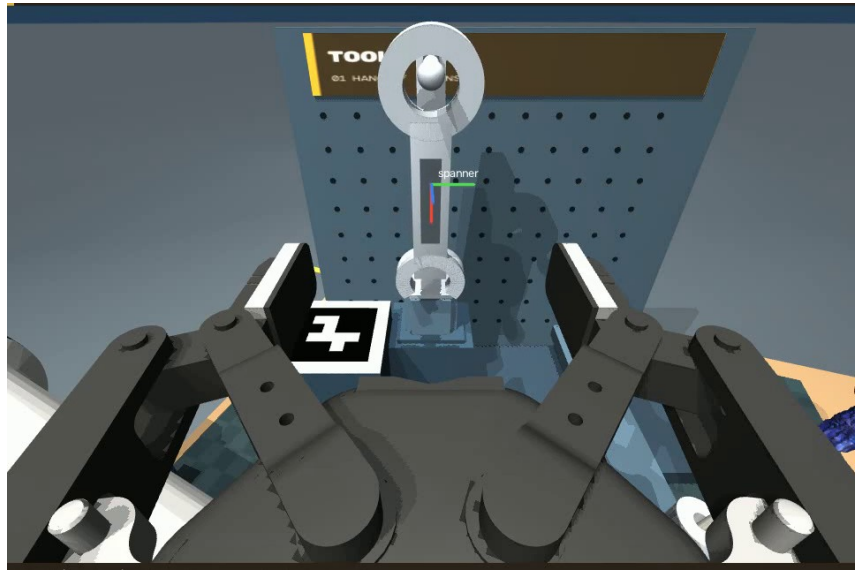
The zoom is from the same original image.

Assumption	Impact on interpretation
Known pickup stations and paired backgrounds	Does not establish arbitrary-clutter recognition
Synthetic scans, ideal FK / depth, rigid mounts	Does not establish real-camera or hardware reliability
Small reused-seed sample	Broader unseen-scene trials remain necessary

Separate development diagnosis; not pooled into the 50-scene localization audit or the 48 robot attempts.

Conclusions and priorities for further experiments

Supported by the evidence	Not established
Independent calibration integrated into visual control	Real-robot accuracy and reliability
Physical four-operation sequence with regrasp	High success rate across arbitrary layouts
Auditable internal feedback comparison	Statistical superiority or external-algorithm ranking



Next experiments

1. Improve pickup planning and bilateral-contact reliability.
2. Test segmentation under changed backgrounds and clutter.
3. Evaluate pose bias and observation scheduling.
4. Use unseen seeds and matched external-estimator baselines.
5. Validate the calibrated pipeline on physical hardware.

References and reproducibility

Appendix

Selected methods

Wen et al. FoundationPose. CVPR, 2024.

Zhang. A flexible new technique for camera calibration. TPAMI, 2000.

OpenCV. Camera and hand–eye calibration documentation.

Chaumette & Hutchinson. Visual servo control, Part I. RAM, 2006.

Todorov et al. MuJoCo. IROS, 2012.

Reviewed alternatives

Labbé et al. MegaPose. arXiv:2212.06870, 2022.

Fang et al. GraspNet-1Billion. CVPR, 2020.

Repository evidence

Calibration images + robot FK

Predicted and evaluation masks

48 complete trial summaries

Original successful and failed replays

12 annotated image plates (PDF / SVG)

Frame indices, crop boxes and source hashes

Full references: docs/LITERATURE_AND_SELECTION.md

Experiment report: visual-feedback-calibrated-v1

Simulation images illustrate observed behavior; quantitative conclusions use all predefined attempted trials.